Derek Chen

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Research Overview

My research focuses on backdrivable robot hardware capable of high-speed environmental interaction, haptic interfaces with transparent torque feedback, and robotic systems for medical procedures. Going forward, I'm interested in building robots that can safely make and withstand physical contact, both with their environment and with the people who interact alongside them.

Education

- **University of California San Diego** *September 2026 - Present*
  M.S. in Electrical Engineering GPA: 3.814/4.00
  - **Relevant Coursework:** Principles of Medical Imaging, Musculoskeletal Health-Injury-and Disease, Robot Manipulation and Control

- **University of California San Diego** *September 2021 - June 2026*
  B.S. in Electrical Engineering - Henry Booker Memorial Honors Award - Provost Honors GPA: 3.700/4.00
  - **Relevant Coursework:** Microwave Systems and Circuits, Analog Integrated Circuit Design, Digital Integrated Circuit Design

Publications

* indicates equal contribution. These papers are also listed at my Google Scholar.

1. **Derek Chen**, Zoe Samuels, Lizzie Peiros, Sujaan Mukherjee, and Michael C. Yip. *"Characterization and Evaluation of Screw-Based Locomotion Across Aquatic, Granular, and Transitional Media."* In: IEEE International Conference on Robotics and Automation (ICRA), 2026.

2. Peihan Zhang, **Derek Chen**, Ishan Duriseti, Florian Richter, Zoe Chiu, Moira Bohley, and Michael C. Yip. *"Dexterous Control of an 11-DOF Redundant Robot for CT-Guided Needle Insertion With Task-Oriented Weighted Policies."* In: IEEE 22nd International Conference on Automation Science and Engineering (CASE), 2026.

3. Elizabeth Peiros*, **Derek Chen***, Lucas Yager*, Edward Lee, Kevin Wang, Zoe Samuels, and Michael C. Yip. *"Physical Twins: Accelerating and Enabling Robot Learning with Phantom Platforms."* In preparation, ACM Transactions on Human-Robot Interaction (T-HRI), 2027.

4. Emma Huang, Moira Bohley, Krishna Jamakhandi, Rebecca Fellbaum, Sophia Tang, Sean Tutton, Albert Hsiao, Alexander Norbash, Changwei Chen, **Derek Chen**, Ananya Rao, Lizzie Peiros, and Michael C. Yip. *"Integrated Respiratory Sensing and Gated Control for Autonomous Needle Insertion Via CT Bore Robotic Arm."* Under review, IEEE International Conference on Robotics and Automation (ICRA), 2027.

5. Elizabeth Peiros, Moira Bohley, Lucas Yager, **Derek Chen**, Rebecca Fellbaum, Chris D'Ambrosia, Ananya Rao, Xiao Liang, and Michael C. Yip. *"Bi-manual Stabilization of the Cervical Spine for Safe Physical Human-Robot Interaction in Rolling Maneuvers."* Under review, IEEE International Conference on Robotics and Automation (ICRA), 2027.

Workshop Papers

1. Elizabeth Peiros*, **Derek Chen***, and Lucas Yager*. *"Physical Twins: Bridging the Gap of Minimal Data and Safe Real-World Human Experiments Enabling AI/RL in Physical Human-Robot Interactions."* In: Workshop on Bridging the Gap between Robot Learning and Human-Robot Interaction, ICRA 2026.

Experience

- **Advanced Robotics and Controls Lab: UCSD** *September 2023 - Present*
  Graduate/Undergraduate Researcher La Jolla, CA

  - **Dynamic Bimanual Arm (video)** *first author, in preparation, Nature Machine Intelligence*
    - Designed and built a bio-inspired, low-inertia, backdrivable 9-DOF bimanual arm, using a spherical parallel manipulator (SPM) to replicate human shoulder rotation and a string-driven wrist to relocate motors closer to the shoulder.
    - Throws objects over 30 ft, achieves 3.6 mm positional repeatability, and withstands repeated hammering impacts.
    - Human-like workspace enables near-egocentric motion retargeting from a human operator.
  - **Haptic Shoulder** *co-first author, Physical Twins (in preparation, T-HRI)*

  - Created a bio-inspired human shoulder haptic device capable of rendering joint limits of able bodied and disabled individuals to ensure adequate representation of all care recipients.
  - Coaxial SPM driven by Maxon motors through capstan cable transmissions minimizes perceived friction, reflecting only motor inertia to the user.
- **CT Bore Robot** *second author, CASE 2026; co-author, ICRA 2027 (under review)*
  - Integrated absolute encoders at the end effector of an 11-DOF CT bore robot, streaming readings over Bluetooth as micro-ROS topics.
  - Reduced the robot's weight to enable mounting on a 6-DOF industrial arm.
  - Mentored students designing a breathing-compensated needle insertion mechanism and a lung phantom.
- **Hall Effect Sensor Board**
  - Designed a 13×13 mm Hall effect sensor board to send 12-bit angle readings over I^2C. Used in multiple projects across the research lab.
- **ARCSnake Testbed** *first author, ICRA 2026*
  - Designed an aquatic module for an amphibious screw-propelled testbed.
  - Led data collection across sand, water, and transitional media.
- **Pressure-Sensing Grasping Claw**
  - Created a novel grasping claw with pressure sensors integrated into the finger pads.
  - Used a compact capstan pulley system at each joint and lightweight twisted string actuators, allowing safe manipulation of human body parts without bruising.
- **Surgical Catheter Controller**
  - Designed a one-handed controller for a custom surgical catheter, enabling precise manipulation of the device.

- **ASML** *June 2024 - September 2024*
  Mechatronics Intern Wilton, CT
  - Designed and sourced PCBs for high vacuum environments following proprietary design constraints to minimize particle outgassing in addition to preserving signal integrity.
  - Created custom fixture to calibrate motor force constants for gain balancing as well as a fixture to measure the run out of a cable duct tester at operational speed.

- **Cadence Design Systems** *June 2023 - September 2023*
  Electrical Engineering Intern: Intellectual Property Group San Jose, CA
  - Engineered a robotic system capable of remote chip testing. Incorporates a custom aluminum sheet metal frame to resist upward pressure generated from the IC cooler, combined with a distinctive rotational and translational motion system, resulting in an eightfold reduction in chip switching time compared to the previous model.
  - Conducted high-speed channel simulations using Sigrity to optimize channel performance (insertion loss, return loss) and ensure compliance with IEEE standards and SerDes chip performance.

- **SEDS at UCSD** *October 2021 - June 2025*
  Rocket Lander Testbed Avionics and Structures, Director of Electrical La Jolla, CA
  - Designed an efficient power system for a self-landing rocket, integrating I^2C-controlled relays, embedded voltage sensors, active battery balancing, and buck-boost converters to ensure safety and provide stable 24VDC/12VDC power to essential vehicle components.
  - Contributed to developing a thrust vectoring system, utilizing custom machined parts with DFM considerations and linear actuators to effectively support and gimbal the in-house Nephas rocket engine.

- **Avicena Tech** *June 2022 - September 2022*
  Packaging Engineering Intern, Mechanical Engineering Consultant Mountain View, CA
  - Designed and developed an automated goniometer to accurately measure and characterize the far-field pattern of LEDs. Utilized RC filtering, lock-in amplifiers, effective shielding, and mechatronics to optimize the device's performance.

SKILLS

- Tools: Eagle, PSpice, SolidWorks, Siemens NX + Teamcenter, Fusion 360, Fusion CAM, Sigrity
- Languages: C, MATLAB, SPICE, SystemVerilog
- Skills: PCB Layout and Design, Circuit Analysis, DFM, Machining, CAD, CAM, Soldering, Path Planning, Kinematics, Gravity Comp/Feedforward Control

SERVICE

- Reviewer, IEEE International Conference on Robotics and Automation (ICRA), 2027.

MENTORSHIP

- 5 students mentored in mechanical design and DFM at ARCLab. Mentees went on to contribute as authors in ICRA submissions. Run electronics and capstan design reviews for the entire ARCLab.
- Over 20 students mentored in basic electrical engineering skills like soldering and simple circuit design at SEDS.
- Over 40 students introduced to robotics with help in both design and manufacturing as a FIRST Robotics mentor.